

REPORT

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Colab link:

<https://colab.research.google.com/drive/14lkvgZyabYTkATKTE4LHrb6HrqPV6gIM?usp=sharing>

1. Title

NLP-Based Sentiment Analysis of YouTube Comments Using Machine Learning

2. Problem Statement

YouTube contains a large number of comments posted by users. These comments express opinions, reactions, feedback, and experiences about videos.

Manually analysing a large number of comments is difficult, time-consuming, and inefficient.

The objective of this project is to develop an automated system that analyses YouTube comments and classifies them into **Positive, Neutral, or Negative** sentiments using **Natural Language Processing (NLP)** and **Machine Learning**.

3. Objectives

The main objectives of this project are:

1. To load and preprocess YouTube comments.
2. To clean textual data using NLP techniques.
3. To remove unnecessary words and perform text normalization.

4. To convert text into numerical features using TF-IDF.
 5. To train a machine learning model for sentiment classification.
 6. To classify comments into Positive, Neutral, and Negative sentiments.
 7. To evaluate the performance of the trained model.
 8. To predict the sentiment of new YouTube comments.
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4. Problem Relevance

YouTube comments provide useful information about how users respond to videos. They may contain positive feedback, negative opinions, questions, suggestions, or neutral statements.

Analysing these comments manually becomes difficult when the number of comments is large. An automated sentiment analysis system can process comments quickly and identify the general sentiment expressed by users.

Why NLP is Used

Natural Language Processing is suitable for this project because the input data consists of human language in the form of YouTube comments.

NLP techniques help to:

- Clean the comments.
- Convert text into a standard form.
- Remove unnecessary words.
- Tokenize the text.
- Reduce words to their meaningful forms.
- Extract useful features from the text.
- Prepare the text for machine learning.

Beneficiaries

The system can be useful for:

- **Content creators** – to understand audience reactions.
- **Businesses** – to analyse customer opinions and feedback.
- **Researchers** – to study public opinions and user behaviour.

- **Online platforms** – to analyse large numbers of comments.

Applications

The project can be applied to:

- YouTube audience analysis
- Opinion mining
- Feedback analysis
- Social media sentiment analysis
- Customer opinion analysis

5. Dataset

5.1 Dataset Name

YouTube Comments Sentiment Dataset

5.2 Dataset File

YoutubeCommentsDataSet.csv

5.3 Dataset Source

Dataset Source: Kaggle

5.4 Dataset Size

The dataset size is obtained from the Google Colab execution.

- **Initial Dataset Size:** 18,408 rows × 2 columns
- **Final Dataset Size:** 18,139 rows × 3 columns
- **Final Number of Rows:** 18,139
- **Final Number of Columns:** 3

5.5 Dataset Attributes

The important fields used in this project are:

Comment: This field contains the text of the YouTube comments.

Sentiment: This field contains the sentiment label associated with each YouTube comment, such as Positive, Neutral, or Negative.

5.6 Target Variable

Target Variable: sentiment

The target variable represents the sentiment category that the model needs to predict.

5.7 Sentiment Classes

The project uses the following sentiment categories:

- **Positive**
- **Neutral**
- **Negative**

The final classes should be confirmed from the actual dataset output.

5.8 Data Preparation

The dataset is prepared using the following steps:

1. Load the CSV dataset.
2. Examine the dataset structure.
3. Check for missing values.
4. Select the Comment and Sentiment columns.
5. Rename the columns as comment and sentiment.
6. Remove missing records.
7. Remove duplicate records.
8. Standardize the sentiment labels.
9. Keep the required sentiment categories.
10. Clean and preprocess the comments.
11. Remove comments that become empty after preprocessing.

5.9 Dataset Limitations

The dataset may contain:

- Informal language.
- Slang.
- Ambiguous comments.
- Different writing styles.
- Unbalanced sentiment categories.
- Comments from a limited range of YouTube topics.

The quality of the dataset can directly affect the performance of the sentiment classification model.

6. Methodology

The methodology used in this project follows the sequence:

YouTube Comments



Data Loading



Data Cleaning



Text Preprocessing



Stopword Removal



Lemmatization



Train-Test Split



TF-IDF Feature Extraction



Logistic Regression



Sentiment Prediction



Model Evaluation

6.1 Data Loading and Cleaning

The dataset is loaded into Google Colab using the Pandas library.

The dataset structure, column names, sample records, and missing values are examined.

Missing records and duplicate records are removed to improve the quality of the data used for training.

6.2 Text Preprocessing

Text preprocessing is the process of cleaning and preparing the YouTube comments before applying machine learning. The following NLP techniques are used:

- **Lowercase conversion** – Converts all text into lowercase to maintain consistency.
- **URL removal** – Removes website links and URLs from the comments.
- **HTML tag removal** – Removes HTML tags that are not useful for sentiment classification.
- **Special-character and number removal** – Removes unnecessary symbols and numbers from the text.
- **Tokenization** – Splits each comment into individual words or tokens.
- **Stopword removal** – Removes common words such as *the*, *is*, *and*, *a* that provide less useful information for classification.
- **Lemmatization** – Converts words into their basic or dictionary form. For example, *running* can be converted to *run*.

6.3 Train-Test Split

The processed dataset is divided into training and testing data.

The project uses:

- **80% training data**
- **20% testing data**

The training data is used to train the Logistic Regression model.

The testing data is used to evaluate how well the trained model performs on unseen comments.

The code uses:

```
random_state = 42
```

and stratification is used to maintain the sentiment distribution between the training and testing sets.

6.4 TF-IDF Feature Extraction

Machine learning algorithms cannot directly process raw text. Therefore, the cleaned comments are converted into numerical features using **TF-IDF (Term Frequency-Inverse Document Frequency)**.

TF-IDF gives importance to words based on how frequently they occur in a comment and how useful they are across the collection of comments.

The project uses the following TF-IDF settings:

- Maximum features: **5000**
- N-gram range: **(1, 2)**
- Minimum document frequency: **2**
- Maximum document frequency: **0.95**
- Sublinear TF scaling: **Enabled**

The use of (1, 2) means that both individual words and two-word combinations are considered.

For example:

amazing

very helpful

can both become TF-IDF features.

6.5 Logistic Regression

Logistic Regression is used as the machine learning classification algorithm.

The model learns the relationship between the TF-IDF features and the sentiment labels.

After training, the model predicts whether an unseen YouTube comment belongs to the Positive, Neutral, or Negative category.

Logistic Regression was selected because it is a simple, understandable, and suitable algorithm for text classification with TF-IDF features.

The assignment emphasizes simple, understandable solutions and requires the implementation to be based on the methodology described in the report.

6.6 Sentiment Prediction

After training the model, a prediction function is created.

The function accepts a new YouTube comment and performs the following steps:

New Comment

↓

Preprocessing

↓

TF-IDF Conversion

↓

Logistic Regression

↓

Predicted Sentiment

The project tests positive, negative, and neutral example comments.

It also provides an interactive option where the user can enter a new YouTube comment and obtain its predicted sentiment.

6.7 Model Evaluation

The trained model is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Classification Report

- Confusion Matrix

These evaluation measures are specifically suitable for classification projects and are listed in the assignment instructions.

7. Implementation

7.1 Programming Language

Python

7.2 Platform

Google Colab

7.3 Libraries Used

The following Python libraries are used:

- **Pandas** – For data loading and data manipulation.
- **NumPy** – For numerical operations.
- **Matplotlib** – For creating graphs.
- **Seaborn** – For data visualization.
- **NLTK** – For NLP preprocessing.
- **Scikit-learn** – For TF-IDF, Logistic Regression, and model evaluation.

7.4 NLP Techniques Used

The following NLP techniques are used:

- Text normalization
- Lowercase conversion
- URL removal
- HTML removal
- Tokenization
- Stopword removal
- Lemmatization
- TF-IDF feature extraction

7.5 Machine Learning Algorithm

Logistic Regression

Logistic Regression is used to classify the processed YouTube comments into Positive, Neutral, and Negative sentiment categories.

7.6 Important Implementation Details

The implementation uses:

- `train_test_split()` for splitting the dataset.
- `TfidfVectorizer()` for feature extraction.
- `LogisticRegression()` for classification.
- `accuracy_score()` for calculating accuracy.
- `precision_score()` for calculating precision.
- `recall_score()` for calculating recall.
- `f1_score()` for calculating F1-score.
- `classification_report()` for detailed classification results.
- `confusion_matrix()` for analysing correct and incorrect predictions.

The source code is organized into separate sections corresponding to the methodology. The code is designed to be readable, understandable, commented, and executable.

8. Results

The trained Logistic Regression model is evaluated using the test dataset.

8.1 Dataset Output

Dataset Shape: (18408, 2)

	Comment	Sentiment
0	lets not forget that apple pay in 2014 require...	neutral
1	here in nz 50 of retailers don't even have con...	negative
2	i will forever acknowledge this channel with t...	positive
3	whenever i go to a place that doesn't take app...	negative
4	apple pay is so convenient secure and easy to ...	positive

8.2 Evaluation Metrics

Accuracy: 0.752205

Precision: 0.737477

Recall: 0.752205

F1-Score: 0.735757

8.3 Classification Report

The classification report provides detailed performance for each sentiment class.

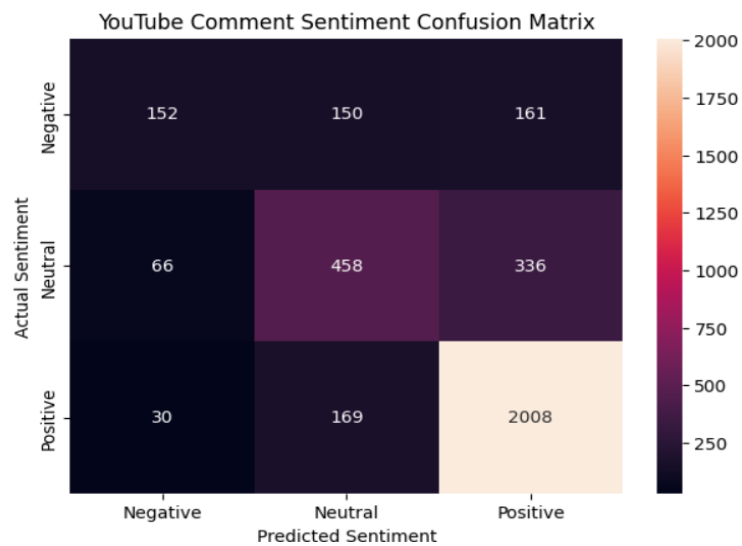
It includes:

- Precision
- Recall
- F1-score
- Support

...	precision	recall	f1-score	support
Negative	0.61	0.33	0.43	463
Neutral	0.59	0.53	0.56	860
Positive	0.80	0.91	0.85	2207
accuracy			0.74	3530
macro avg	0.67	0.59	0.61	3530
weighted avg	0.73	0.74	0.73	3530

8.4 Confusion Matrix

The confusion matrix shows the number of comments that were correctly and incorrectly classified into the three sentiment categories.

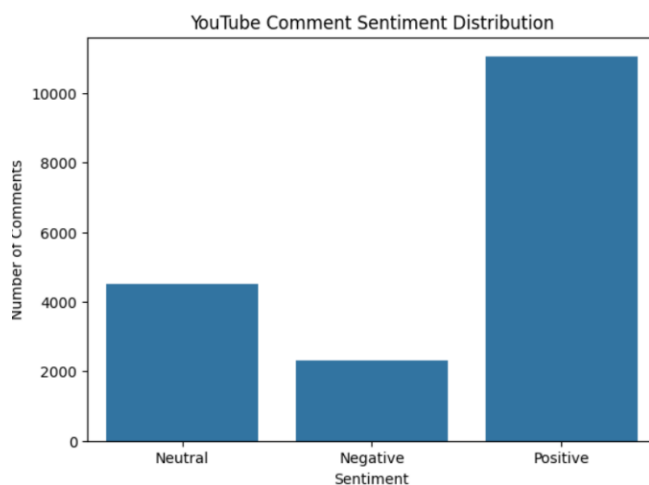


The classes are:

- Negative
- Neutral
- Positive

8.5 Sentiment Distribution

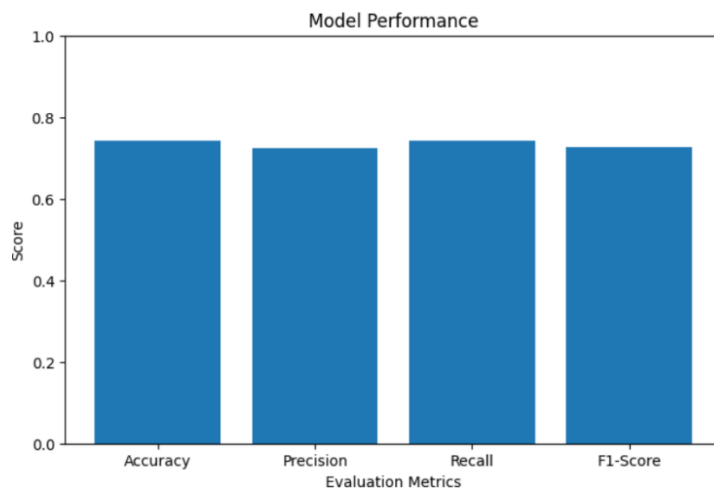
The sentiment distribution graph shows the number of comments belonging to each sentiment category.



8.6 Model Performance Graph

The model performance graph compares:

- Accuracy
- Precision
- Recall
- F1-Score



8.7 Results Analysis

The model performance is analysed using the evaluation metrics.

Accuracy represents the overall proportion of correctly classified comments.

Precision indicates how accurately the model identifies comments belonging to a particular sentiment category.

Recall indicates how effectively the model identifies comments belonging to the relevant sentiment category.

F1-Score provides a balance between precision and recall.

The confusion matrix provides additional information about which sentiment classes are correctly classified and which classes are confused with one another.

9. Limitations

The project has the following limitations:

1. The model may have difficulty understanding sarcasm.

2. Slang and informal language may affect predictions.
 3. Mixed-language comments may be difficult to classify.
 4. Emojis and special expressions may lose some information during preprocessing.
 5. Some comments may have ambiguous meanings.
 6. Model performance depends on the quality and distribution of the dataset.
 7. The model may not correctly classify all unseen comments.
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10. Future Scope

The project can be improved in the future through:

1. Using a larger and more diverse YouTube comments dataset.
 2. Supporting multiple languages.
 3. Improving preprocessing for slang, emojis, and informal expressions.
 4. Using advanced NLP and deep learning models.
 5. Developing a real-time YouTube comment sentiment analysis system.
 6. Developing a web application for sentiment prediction.
 7. Comparing multiple machine learning algorithms to identify the best-performing model.
 8. Improving evaluation using a wider range of test data.
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11. Conclusion

This project developed an NLP-based sentiment analysis system for YouTube comments.

The comments were cleaned and preprocessed using NLP techniques including lowercase conversion, URL removal, HTML tag removal, tokenization, stopwords removal, and lemmatization.

TF-IDF was used to convert the processed comments into numerical features. Logistic Regression was then used to classify the comments into Positive, Neutral, and Negative sentiment categories.

The model was evaluated using Accuracy, Precision, Recall, F1-Score, Classification Report, and Confusion Matrix.

The project demonstrates how Natural Language Processing and Machine Learning can be used to automatically analyse opinions expressed in YouTube comments.

The project also provides a foundation for future improvements such as larger datasets, multilingual support, advanced NLP models, and real-time sentiment analysis.